La Consolacion University Philippines
Valenzuela St., Capitol View Park Subdivision, Bulihan, City of Malolos, Bulacan

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SCHOOL YEAR 2025-2026
COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. COURSE CODE : IE-403

II. COURSE TITLE : Operation Management

III. COURSE DESCRIPTION :

This course provides students with a comprehensive understanding of the principles, strategies, and tools used in managing operations within manufacturing and service organizations. It explores key topics such as operations strategy, process design, capacity planning, supply chain management, inventory control, quality assurance, and project scheduling. Emphasis is placed on applying quantitative and qualitative methods to solve real-world operational challenges, improve productivity, and enhance organizational competitiveness. Through case studies, simulations, and team projects, students will develop critical thinking and decision-making skills essential for effective operations leadership in dynamic business environments.

Anchored in Marian and Augustinian values, this course introduces students to the principles and tools of managing operations in manufacturing and service organizations. It emphasizes strategic thinking, process design, and data-driven decision-making to solve real-world challenges and improve productivity in ethical and socially responsive ways.

IV. PRE-REQUISITE :

V. CREDIT UNITS : 3 units lecture

VI. COURSE LEARNING OUTCOMES:

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;*

	● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker **EGA 2:** Competent Catholic Augustinian-Marian Professional **EGA 3:** Socially Responsive Steward **EGA 4:** Transformative Lifelong learner	**PO1.** Ability to apply knowledge of mathematics and science to solve complex engineering problems; **PO2.** Ability to design and conduct experiments, as well as to analyze and interpret data. **PO3.** Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards. **PO4.** Ability to function on multidisciplinary teams; **PO5.** Ability to identify; formulate, and solve complex engineering problems. **PO6.** Understanding of professional and ethical responsibility. **PO7.** Ability to communicate effectively. **PO8.** Broad education necessary to understand the impact of engineering	**CLO1:** Apply interdisciplinary knowledge from computing, mathematics, physical and social sciences to identify, formulate, and solve complex engineering problems through the design and execution of experiments, and the development of systems or processes that meet real-world needs within economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability constraints, in accordance with professional standards. (EGA 1) (PO 1, 2, 3, 5) **CLO2:** Exhibit professional and ethical responsibility while communicating effectively and applying engineering and management principles to lead

	solutions in a global, economic, environmental, and societal context. **PO9.** Recognition of the need for, and an ability to engage in life-long learning; **PO10.** Knowledge of contemporary issues. **PO11.** Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and **PO12.** Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments. **PO13:** Ability to design, develop, implement and improve integrated system that include people materials, implementation, equipment and energy.	and collaborate within multidisciplinary teams, guided by Catholic Augustinian-Marian values. (EGA2) (PO 6, 7, 12) **CLO3:** Collaborate effectively in multidisciplinary teams to develop engineering solutions that address global, economic, environmental, and societal challenges, while demonstrating a commitment to lifelong learning and socially responsible stewardship. (EGA 3) (PO 4, 8, 9) **CLO4:** Apply modern engineering tools and techniques to design and improve integrated systems involving people, materials, equipment, and energy, while demonstrating awareness of contemporary issues and a commitment to lifelong learning in pursuit of transformative professional growth. (EGA 4) (PO 10, 11, 13)

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week1	ILO 1. Identify real-world operational problems and explain how mathematics and science can be applied to address them. (CLO 1) ILO 2. Describe the role of ethics and professionalism in operations decision-making. (CLO 2) ILO 3. Explain the societal and environmental impact of operations decisions. (CLO 3) ILO 4. Identify contemporary tools and technologies used in operations management. (CLO 4)	I. Introduction to Operations Management A. Definition, scope, and importance of OM B. Historical development C. Role of operations in the organization	• Lecture-discussion on OM concepts • Class icebreaker: "How Operations Affect You" • Short video: "What is Operations Management?" • Reflection paper	• Diagnostic quiz; reflection paper on "How Operations Affect Daily Life"	• Discipline: Encourages students to value order and systematic processes as a foundation of effective operations. • Integrity: Builds awareness of honesty and ethical behavior as essential in managing operational activities and decision-making. • Responsibility: Instills accountability for roles and decisions within an organization's operational system.

Week 2-3	ILO1. Formulate problem statements and propose analytical models using engineering principles. Design experiments or simulations to test operational hypotheses. (CLO 1) ILO 2. Communicate operational goals and constraints clearly in written and oral formats. Participate in team-based discussions and contribute constructively to group tasks. (CLO 2) ILO 3. Identify roles and responsibilities in a multidisciplinary team setting. Engage in collaborative problem-solving with peers from diverse backgrounds. (CLO 3) ILO 4. Use basic engineering software to model operational processes.	II. Operations Strategy and Competitiveness • Relationship between operations and corporate strategy • Productivity and performance measures III. Product and Service Design • Product life cycle • Design for manufacturability and serviceability • Value engineering	• Group discussion on business competitiveness • Case analysis: "Operations Strategy of Toyota" • Productivity computation exercise • Lecture and examples of design innovation • Service blueprinting activity (group) • Mini-presentation of redesigned products	• Short quiz; group analysis report on business competitiveness • Group output: product redesign proposal with brief rationale	• **Excellence, Accountability, Creativity, Innovation, Customer Orientation:** Encourages students to pursue excellence and accountability in developing effective operations strategies while fostering creativity and innovation in designing customer-centered products and services.

	Apply digital tools to visualize system layouts and workflows. (CLO 4)				
Week 4-5	ILO 1. Analyze and interpret data from operations scenarios to draw evidence-based conclusions. Apply optimization techniques to improve system performance under realistic constraints. (CLO1) ILO 2. Apply leadership principles to guide team decision-making in project settings. Resolve conflicts and negotiate solutions within multidisciplinary teams. (CLO 2) ILO 3. Apply global and local perspectives to operational challenges. Reflect on the importance of lifelong learning in engineering practice. (CLO 3)	IV. Process and Capacity Planning • Types of processes (job, batch, mass, continuous) • Process selection and analysis • Capacity planning concepts V. Facility Layout and Location Planning • Factors affecting location • Types of layout (process, product, cellular, fixed position) • Relationship between layout and productivity	• Lecture with process flow diagrams • Problem-solving: capacity planning scenarios • Simulation: “Bottleneck Identification” • Classroom layout design exercise • Site location case analysis • Quiz #1	• Problem set on capacity planning; short reflection on process design • Layout design sketch submission; quiz on location factors	• **Discipline, Responsibility, Efficiency:** Encourages systematic planning, accountability, and effective resource utilization in process and facility design.

	ILO 4. Analyze system performance using simulation software. Design integrated systems that include people, materials, and equipment. (CLO 4)				
Week 6		**Preliminary Examination**	**Exam**		
Week 7-9	ILO 1. Evaluate alternative solutions using trade-off analysis and decision matrices. Use mathematical tools to model process flows and resource allocation. Solve complex scheduling or inventory problems using quantitative methods. (CLO 1) ILO 2. Demonstrate ethical reasoning in evaluating operational trade-offs. Prepare and deliver a presentation on an operations case study. Collaborate with peers to develop a shared project plan. (CLO 2)	VI. Forecasting in Operations • Qualitative and quantitative forecasting methods • Time series models • Forecast accuracy and error measures VII. Work Design and Measurement • Job design principles • Methods analysis • Time and motion studies VIII. Inventory Management • Types and functions of inventory • EOQ model	• Lecture-demo on forecasting methods • Spreadsheet exercise: moving average & exponential smoothing • Short quiz • Motion study video analysis • Group activity: Work simplification project • Reflection: "Ergonomics in Workplaces" • Lecture-demo: EOQ computation • Case: "Balancing Inventory Costs" • Worksheet activity using sample data	• Computed worksheet; short quiz on forecasting accuracy • Group output: work simplification report; oral presentation • Problem-solving worksheet; quiz on EOQ	• **Accuracy, Integrity, Accountability, Professionalism:** Reinforces ethical and precise decision-making, responsibility, and professionalism in forecasting, work systems, and inventory control.

	ILO 3. Analyze operations scenarios with consideration for social and environmental outcomes. Participate in team-based sustainability assessments. Integrate feedback from multiple disciplines into project development. (CLO 3) ILO 4. Implement productivity tools to streamline operations tasks. Evaluate the effectiveness of modern tools in solving operational problems. Use data analytics platforms to interpret operational metrics. (CLO 4)	• Inventory control systems (ABC, JIT)			
Week 10-11	ILO 1. Interpret system behavior through simulation and data visualization. Apply scientific reasoning to assess reliability and risk in operations. (CLO 1)	IX. Supply Chain Management • Components and flow of supply chain • Supplier management and logistics • Supply chain coordination	• Role-playing: Supply chain coordination game • Guest speaker (if available) from logistics industry • Reflection journal	• Written reflection; group case study presentation • Quality improvement plan; group output and short quiz	• **Collaboration, Integrity, Excellence, Commitment:** Promotes teamwork, ethical practices, and dedication to achieving high-quality standards

	ILO 2. Document team processes and decisions using professional standards. Reflect on personal leadership style and team contributions. (CLO 2) ILO 3. Explore contemporary issues affecting operations and supply chains. Demonstrate adaptability and openness to new ideas in team settings. (CLO 3) ILO 4. Apply engineering tools to optimize resource utilization. Integrate automation and digital solutions into system design. (CLO 4)	X. Quality Management • Total Quality Management (TQM) • Six Sigma and continuous improvement • Statistical Process Control (SPC)	• Lecture: Quality frameworks • Group workshop: Fishbone and Pareto analysis • Quality improvement mini-project		across supply chain operations.
Week 12		**Midterm Examination**	**Exam**		
Week 13-14	ILO 1. Design integrated systems that meet operational goals within economic and environmental constraints.	XI. Scheduling and Project Management • Scheduling methods and sequencing • Gantt Charts, CPM, and PERT	• Lecture-demo: Gantt chart, CPM/PERT computation • Group project scheduling simulation	• CPM/PERT problem set; short quiz • Group reflection report; photo documentation of 5S activity	• **Punctuality, Teamwork, Efficiency, Continuous Improvement:** Emphasizes time management,

	Improve existing processes using engineering tools and sustainability principles. (CLO 1) ILO 2. Lead a team in developing a sustainable operations proposal. Communicate technical findings to diverse audiences. (CLO 2) ILO 3. Propose socially responsive solutions to operational problems. Collaborate on a community-based operations improvement project. (CLO 3) ILO 4. Develop a tool-supported plan for improving an integrated system. Test and refine system components using engineering software. (CLO 4)	• Time-cost trade-offs XII. Lean Systems and Just-in-Time (JIT) • Principles of lean operations • Waste elimination • 5S and Kaizen	• Quiz #2 • 5S classroom audit • Lean simulation exercise (“Paper Airplane Factory”) • Group reflection on continuous improvement		collaboration, and a commitment to improving processes and eliminating waste.
Week 15-17	ILO 1. Justify design decisions using data analysis and performance metrics.	XIII. Maintenance and Reliability	• Case study: Equipment downtime problem	• Case analysis paper; short quiz • Group presentation:	• Dependability, Adaptability, Innovation, Environmental Stewardship:

	Synthesize knowledge from multiple domains to propose innovative operational solutions. Present a comprehensive system design that solves a complex operational challenge. (CLO 1) ILO 2. Evaluate team performance and suggest improvements. Apply project management tools to coordinate team tasks. Defend team-based project outcomes in a formal presentation. (CLO 2) ILO 3. Evaluate the long-term societal impact of proposed systems. Reflect on personal growth and learning throughout the course. Present a socially responsible operations plan developed by a multidisciplinary team. (CLO 3)	• Preventive and predictive maintenance • Reliability analysis • Equipment effectiveness XIV. Operations Technology and Innovation • Automation and Industry 4.0 • ERP systems • Data-driven operations XV. Sustainability and Green Operations • Environmental management • Sustainable supply chains • Circular economy principles	• Group presentation on reliability strategies • Lecture and video clips: “Industry 4.0 in Action” • Mini research on local firms using ERP • Group sharing • Case analysis: “Green Manufacturing Practices” • Group project: Sustainable Operations Plan presentation	“Technology in Operations” • Final project submission: Sustainable Operations Plan	Encourages reliability, openness to technological change, creative problem-solving, and responsibility toward sustainable operations.

	ILO 4. Assess the sustainability of tool-based system improvements. Explore emerging technologies and their applications in operations. Demonstrate proficiency in using modern tools during final project presentation. (CLO 4)				
Week 18		**FINAL EXAMINATION**	Exam week		

VIII. COURSE REQUIREMENTS :

1. Term examinations
2. Quizzes
3. Seatwork, assignment and/or project
4. Class attendance
5. Class participation

IX. REFERENCES:

A. BOOKS

Heizer, J., Render, B., & Munson, C. (2023). Operations management (14th ed.). Pearson Education.

Stevenson, W. J. (2021). Operations management (14th ed.). McGraw-Hill Education.

Krajewski, L. J., Malhotra, M. K., & Ritzman, L. P. (2019). Operations management: Processes and supply chains (12th ed.). Pearson Education.

Russell, R. S., & Taylor, B. W. (2019). Operations and supply chain management (10th ed.). John Wiley & Sons.

Slack, N., Brandon-Jones, A., & Burgess, N. (2022). Operations management (10th ed.). Pearson Education.

X. GRADING SYSTEM :

1. Class Standing 60%
   - Quiz
   - Recitation
   - Attendance
2. Examination 40 %
3. Term Grade CS (60%) + TE (40%)
4. Final Rating Prelim Grade + Midterm Grade + Final Grade

   3
5. Passing Grade: 50%

Prepared by:	Checked and Reviewed by:	Approved by:
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Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	